

Algebra II with Trigonometry

Chapter 5.4

Olympiad Solutions

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[4401270_alg-2trig-h-chp-5-4-note-docx.pdf](#)

Solutions

1. Compute the exact value of $\arccos(\sin(\arctan \frac{3}{4}))$.

Step 1: Let $\theta = \arctan \frac{3}{4}$.

Draw a right triangle with legs 3 (opposite) and 4 (adjacent). Then,

$$\sin \theta = \frac{3}{5}$$

where $5 = \sqrt{3^2 + 4^2}$.

Step 2: Compute $\arccos(\sin \theta)$:

$$\arccos\left(\sin\left(\arctan \frac{3}{4}\right)\right) = \arccos\left(\frac{3}{5}\right)$$

But, $\theta = \arctan \frac{3}{4}$ is in $(0, \pi/2)$, so $\sin \theta > 0$ and $\arccos(\sin \theta)$ is in $[0, \pi]$.

Let $\alpha = \arccos(\sin \theta)$, so $\cos \alpha = \sin \theta$.

But observe:

$$\sin \theta = \sin(\pi/2 - \theta) = \cos \theta$$

But with our triangle, $\cos \theta = \frac{4}{5}$, which is not $\frac{3}{5}$. Thus, not the same.

Instead, proceed:

Let $\phi = \arcsin\left(\frac{3}{5}\right) \implies \phi \in (0, \pi/2)$. Then,

$$\arccos\left(\frac{3}{5}\right) + \arcsin\left(\frac{3}{5}\right) = \frac{\pi}{2}$$

So,

$$\arccos\left(\sin\left(\arctan\frac{3}{4}\right)\right) = \arccos\left(\frac{3}{5}\right) = \frac{\pi}{2} - \arcsin\left(\frac{3}{5}\right)$$

But $\arcsin\left(\frac{3}{5}\right) = \arctan\left(\frac{3}{4}\right)$, since in the reference triangle, opposite/adjacent = 3/4.

Thus,

$$\boxed{\arccos\left(\sin\left(\arctan\frac{3}{4}\right)\right) = \frac{\pi}{2} - \arctan\frac{3}{4}}$$

Or, explicitly, $\arccos(\sin(\arctan(3/4))) = \arctan\left(\frac{4}{3}\right)$, since

$$\tan\left(\frac{\pi}{2} - x\right) = \cot x,$$

and $\cot(\arctan(3/4)) = 4/3$, so $\arctan(4/3)$.

Final Answer:

$$\boxed{\arctan\left(\frac{4}{3}\right)}$$

2. Given that $\theta = \arcsin\left(\frac{5}{13}\right)$, find $\cot(2\theta)$.

Step 1: We have $\sin \theta = \frac{5}{13}$.

$$\text{So, } \cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \left(\frac{25}{169}\right)} = \sqrt{\frac{144}{169}} = \frac{12}{13}.$$

Step 2: Compute $\tan \theta$:

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{5}{12}$$

Step 3: Double angle formula for cotangent:

$$\cot(2\theta) = \frac{1 - \tan^2 \theta}{2 \tan \theta}$$

Plug in $\tan \theta = 5/12$:

$$\cot(2\theta) = \frac{1 - (5/12)^2}{2 \times (5/12)} = \frac{1 - 25/144}{10/12} = \frac{119/144}{10/12} = \frac{119}{144} \cdot \frac{12}{10} = \frac{119 \cdot 12}{144 \cdot 10}$$

Final Answer:

$$\boxed{\frac{119}{120}}$$

3. If $\arctan x + \arctan y = \frac{\pi}{4}$, what is the value of $x + y + xy$?

Step 1: Recall

$$\arctan x + \arctan y = \arctan \left(\frac{x + y}{1 - xy} \right) \quad (\text{if } xy < 1)$$

Set equal to $\pi/4$:

$$\arctan \left(\frac{x + y}{1 - xy} \right) = \frac{\pi}{4} \implies \frac{x + y}{1 - xy} = \tan \frac{\pi}{4} = 1$$

So,

$$x + y = 1 - xy \implies x + y + xy = 1$$

Final Answer:

$$\boxed{1}$$

4. Let $a = \arccos\left(\frac{1}{3}\right)$ and $b = \arcsin\left(\frac{2\sqrt{2}}{3}\right)$. Without a calculator, determine whether $a = b$.

Let's check if $\cos a = \sin b$ or if the numbers are related.

Step 1: Note that $\arcsin t + \arccos t = \frac{\pi}{2}$, so unless $b = \frac{\pi}{2} - a$, they are not equal.

Step 2: Compare $\cos a = \frac{1}{3}$, $\sin b = \frac{2\sqrt{2}}{3}$.

Set $a = b$ and see if $\cos a = \sin a$:

$$\cos a = \frac{1}{3}$$

$$\sin a = \sqrt{1 - \frac{1}{9}} = \sqrt{\frac{8}{9}} = \frac{2\sqrt{2}}{3}$$

So $\sin a = \frac{2\sqrt{2}}{3}$.

Thus,

$$b = \arcsin\left(\frac{2\sqrt{2}}{3}\right)$$

But $a = \arccos\left(\frac{1}{3}\right)$.

But since $\sin a = \sin b$ and both $a, b \in [0, \pi/2]$, and since sine is injective on $[0, \pi/2]$:

Thus, $a = b$.

Final Answer:

$$\boxed{a = b}$$

5. Evaluate $\sin\left(\arccos\left(\frac{x}{\sqrt{1+x^2}}\right)\right)$ in terms of x .

Let $\theta = \arccos\left(\frac{x}{\sqrt{1+x^2}}\right)$, so $\cos \theta = \frac{x}{\sqrt{1+x^2}}$.

$$\sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - \frac{x^2}{1+x^2}} = \sqrt{\frac{1+x^2-x^2}{1+x^2}} = \sqrt{\frac{1}{1+x^2}} = \frac{1}{\sqrt{1+x^2}}$$

Final Answer:

$$\boxed{\frac{1}{\sqrt{1+x^2}}}$$

6. Find all real x such that $\arcsin x + \arccos x = \arctan(2x)$.

Step 1: For all $x \in [-1, 1]$,

$$\arcsin x + \arccos x = \frac{\pi}{2}$$

So,

$$\frac{\pi}{2} = \arctan(2x) \implies 2x \rightarrow \infty \implies x \rightarrow +\infty$$

But $x \leq 1$. So no real x in $[-1, 1]$.

But let's check: are there any real x outside this interval for which both sides are defined? For $x \notin [-1, 1]$, $\arcsin x$, $\arccos x$ are not real.

Thus, **no real solutions**.

Final Answer:

No real solutions for x .